Introduction to Probability

Not-so-bad stuff

Combinatorics, DRVs (Bernoulli, Binomial, Poisson, Geometric, Negative Binomial, Hypergeometric), CRVs (Uniform, Exponential, Normal), expectation, moments, variance. ### Expectation Often, when calculate expectation of annoying thing, use linearity of expectation (it works in spite of non-independence). So often just create a ton of indicator variables (often identically distributed) and sum over them. (Where each indicator variable's probability is nigh-trivial to compute)

Marginal and Joint Distributions

Independence, Covariance, Correlation

Limit Theorems and Inequalities

Tailedness Inequalities

Name	Conditions	Formula
Markov	RV $X \geq 0, \mathbf{E}[X] \in \mathbb{R}$	$P[X \geq a] \leq \frac{\mathbf{E}[X]}{a}$
Chebyshev	$\mathbf{E}[X], \mathbf{V}[X] \in \mathbb{R}$	$P[\|X - \mathbf{E}[X]\| \geq a] \leq \frac{\mathbf{V}[X]}{a^2}$

Limit Theorems

Let $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ where X_i are i.i.d. RVs.

Name	What	Theorem
Weak Law of Large Numbers (WLLN)	Converges to	$\lim_{n \to \infty} \mathbf{P}[\| \overline{X}_n - \mu \| > \varepsilon] = 0$
Strong Law of Large Numbers (SLLN)	Converges to (non-exam.)	$\mathbf{P}\left[\lim_{n \to \infty} \overline{X}_n = \mu\right] = 1$
Central Limit Theorem (CLT)	How it converges	$\sqrt{n} \cdot \frac{\overline{X}_n - \mu}{\sigma} \sim N(0, 1)$

*I sometimes call them SLoN/WLoN
**Technically $\lim_{n \to \infty} F_{Z_n}(a) = \Phi(a)$